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Shaunak Joshi

Battery Modeling Researcher

LinkedIn: Shaunak Joshi
Stanford, CA

Ph.D. candidate specializing in mechanically informed battery modeling across particle, electrode, and cell scales. Experienced in developing physics-based/ML models to quantify battery aging for battery design and advanced BMS applications. Seeking full-time roles in **Battery Modeling**, **Computational Modeling**, and **Multiphysics Simulation**.

Education

Stanford University | Advisors – Prof. Daniel Tartakovsky, Prof. Hamdi Tchelepi — **Sep 2022 – Jun 2027** ***[Expected]***
Doctor of Philosophy, Energy Science & Engineering — GPA - 3.98/4.0

Indian Institute of Technology (IIT) Bombay, India | Advisor - Prof. Venkat Ramadesigan — **Jul 2017 – Jun 2022**
Bachelor & Master of Technology, Energy Science & Engineering — GPA – 8.81/10.0

Work Experience

Computational Materials Scientist Intern, Bosch USA — **Summer 2026**
Multiscale modeling team – Humidity-induced mechanical properties of silicon oxide — *Greater Boston, MA*

- Delivered a thermodynamic wetting model (Python) explaining operational drift in precision MEMS microstructures
- Quantified surface roughness effects, with up to 40% lower adsorption-induced stresses than smooth SiO_x surfaces
- Modeled humidity-dependent unwanted stress development by integrating AFM surface roughness measurements

Battery Modeling Intern, Nissan North America — **Summer 2024**
Materials Informatics (MI) team – Crack propagation-based degradation modeling — *Santa Clara, CA*

- Delivered a validated simulator reducing electrode cracking by 50%, guiding high-V cathode synthesis optimization
- Built a physics-based crack propagation model (COMSOL) with diffusion-induced stresses & cohesive zone mechanics
- Identified inter-granular cracking as the dominant degradation mode by linking simulation results with experiments
- Guided optimal electrode design using accurate surface area growth estimates (~200%) in NMC811 cathodes at >4.3 V

Ph.D. Research – Battery Management Systems (BMS)

Physics-guided ML for real-time battery health prediction in silicon/graphite anodes — **Sep 2026 – Present**
Ph.D. Project 5 – Collaborative work with Prof. Simona Onori, Stanford-SLAC Battery Center — *Stanford, CA*

- Lead author of a research proposal to Precourt Institute of Energy, Stanford on Si/Gr composite anode aging
- Aging studies on commercial Si-Gr cells aimed at studying the effect of Si-phase utilization on Gr-phase degradation
- Next - measuring DMs via cell teardown, P2D degradation model development (COBRAPRO) & FNO-based surrogates

Chemo-mechanical applications in Battery Management Systems (BMS) using neural operators — **Spring 2025**
Course Project | CS 229 – Machine Learning — *Stanford, CA*

- Trained Fourier Neural Operators (FNOs) on simulated multiphysics model data enabling their real-time execution
- Achieved >99% accuracy on diverse chemo-mechanical datasets and quantified model foresight limitations
- Enabled advanced BMS algorithms with stress-based mechanical safety checks at minimal computational cost

Ph.D. Research – Mechanically Informed Microstructure Optimization

Modeling interparticle contact and particle-binder delamination in battery microstructures — **Aug 2025 – Present**
Ph.D. Project 4 – Extending chemo-mechanics to microstructure-level simulations — *Stanford, CA*

- Simulated particle-binder delamination using diffusion-induced stresses and cohesive zone models (CZMs) in Abaqus
- Constructed multi-particle chemo-mechanical models with interaction stresses using open-source simulator GReS
- Showed that contact stresses dominate diffusion-induced stresses by >200% in contacting spherical Si particles

Optimizing particle porosity for mechanical stability in high-capacity electrodes — **Oct 2024 – Jul 2025**
Ph.D. Project 2 – Simulation study guiding electrode design by controlling particle-level porosity — *Stanford, CA*

- Designed a multiphysics model for porous-Si particles, combining finite deformation theory and theory of mixtures
- Introducing 20%/50% porosity results in 30%/70% reduced stresses and 13%/36% reduced volume expansion
- Stability curves help optimize microstructure design to maximize energy density while minimizing degradation

Chemo-mechanical modeling of deforming hollow silicon anode particles **Apr 2023 – Jun 2024**
Ph.D. Project 1 – Modifying single particle models for high volume expansion electrode materials *Stanford, CA*

- Developed a chemo-mechanical model with accurate volume expansion and stress predictions for hollow particles
- An exponential increase in stability as the particle size is reduced upon incorporating effects of deforming volume
- Identified the elastic operation range for optimizing design parameters (size/thickness) to minimize cell aging

Research – Cell Level (P2D) Battery Modeling

Incorporating chemo-mechanics and deforming volume principles in cell-level models **Feb 2025 – Jun 2025**
Ph.D. Project 3 – Collaborative work with Prof. Venkat Subramanian, University of Texas at Austin *Stanford, CA*

- Achieved a speedup of 100× over finite-difference for mechanically-informed SPMs using orthogonal collocation
- Incorporated deforming volume chemo-mechanical SPMs in P2D using Maple at a marginal computational overhead
- The developed technology enables estimating mechanical states for SoH estimation in advanced BMS algorithms

Physics-based mathematical modeling of lithium-sulfur (Li-S) battery cells **Jun 2021 - Jun 2022**
Master's thesis, Advisor: Prof. Venkat Ramadesigan, IIT Bombay *Mumbai, India*

- Implemented a cell-level 1-D Li-S battery model (Maple), modifying P2D model equations for conversion mechanism
- Numerically explained phenomenological polysulfide precipitation effect on the plateauing trend of voltage curves
- Replicated reduced-order 0-D and 1-D Li-S battery cell models on PyBaMM, contributing to its open-source repository

Skills and Coursework

Programming & Tools	COMSOL, MATLAB, Python, PyBaMM, C++, Maple, Ansys, PyTorch, GReS, ParaView, GMSH
Quantitative Research	Physics-based Modeling, Chemo-Mechanical modeling, Computational Solid Mechanics
Mathematical Modeling	Finite Element Analysis, Analytical & Numerical solutions to PDEs, Optimization
Energy	Fluid Mechanics, Solid Mechanics, Nuclear Reactor Theory, Energy Policy, Sustainability
Electrical Systems	Power Electronics, Electrical Machines, Electrical Energy Systems, RE Integration
Batteries	Electrochemistry, Energy Storage Technologies, Corrosion, Battery Modeling & Estimation

First-Author Publications

- **Joshi, S. A.**, Tchelepi, H. A., & Tartakovsky, D. M. (2024). Design of Stable Hollow Particles for Silicon Anodes. ACS Energy Letters, 9(10), 4810-4816.
- **Joshi, S. A.**, Tchelepi, H. A., & Tartakovsky, D. M. (2025). Optimized Particle Porosity to Alleviate Diffusion-Induced Stresses in High-Capacity Electrodes. Journal of The Electrochemical Society, 172(12), 120531.
- **Joshi, S. A.**, Tchelepi, H. A., Tartakovsky, D. M., & Subramanian, V. R. (2026). Efficient reformulation of chemo-mechanical models of high volume-expansion electrode materials in lithium-ion batteries using symmetric polynomials *(in preparation)*

Academic Projects

Residual RL for actuator-bias compensation **Winter 2026**
Course Project | CS 234 - Reinforcement learning *Stanford, CA*

- Developed Soft Actor-Critic (SAC) residual RL to compensate actuator bias in a 7-DoF Franka Panda arm (MuJoCo)
- Achieved a peak 20-episode average of 90% reduction in object-to-target as compared to near 0% for pure RL
- Augmented inverse-kinematics with learned joint-angle corrections to compensate actuator bias in contact-rich tasks

Development of fully implicit 2-D reservoir simulator with wells **Winter 2023**
Course Project | ENERGY 223 - Subsurface Flow Simulation *Stanford, CA*

- Built from scratch a Newton-Raphson solver-based simulator on MATLAB for a two-phase, 2-D transient oil reservoir
- Incorporated automatic time-step control, injection/production wells, and gravitational effects into the 2-D simulator
- Manually constructed the Jacobian matrix to solve the coupled mass-balance equations using finite volume method
- Conducted sensitivity analyses for well locations & mobilities to optimize the total oil production for given constraints

Academic Recognition

- Reviewer | Electrochemical Energy Reviews (IF=36.3), Journal of Computational Physics (IF=3.8)
- Awarded Institute Academic Prize (41 out of 900 students) for exemplary academic performance in AY 2020-21